

Dongda Li

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PhD Researcher in Vision-Language Models and Multimodal Learning

Education

Syracuse University

Expected May 2028

PhD in Computer & Information Science & Engineering; Advisor: [Senem Velipasalar](#)

Xi'an University of Technology

B.E. in Automation, 2015

Research Experience

PhD Researcher

Aug. 2023–Present

Syracuse University

Syracuse, NY

- Study compositional reasoning and multimodal robustness in vision-language models, emphasizing role binding, evidence-grounded image-text evaluation, and reliable readouts.
- Analyze CLIP, SigLIP 2, and related frozen dual encoders to identify where role information remains accessible in regional features and where it is lost by deployed similarity scoring.
- Build controlled probing and evaluation pipelines across SugarCrepe, SugarCrepe++, Winoground, synthetic spatial pairs, and real-image role-binding data.
- Design efficient local-evidence architectures, including crop-based and self-attention readouts, and evaluate computational and accuracy tradeoffs through controlled ablations.
- Audit reward models and VLM-based robot-policy evaluators on **5,106 real-robot episodes** against **3,284 double-blind human comparisons**, analyzing ranking agreement, uncertainty, and implementation sensitivity.

Earlier Research Appointments

Guangzhou University

2018–2022

Research Assistant

Cyberspace Institute of Advanced Technology

- Reinforcement learning, neural combinatorial optimization, meta-learning, and learning-based combinatorial decision making.

The University of Hong Kong

2017–2018

Research Assistant

Department of Computer Science

- Robotic networks, wireless sensor networks, and UAV-enabled sensing.

Beijing Institute of Technology

2016–2017

Research Assistant

UAV Autonomous Control Institute

- UAV control, navigation, and data fusion; [core member of the MBZIRC 2017 champion team](#).

Selected Manuscripts

1. **Dongda Li et al.** “The Readout, Not the Representation: Why Frozen Dual-Encoder Models Fail at Role Binding.” *Under review at TMLR*, 2026. Controlled three-backbone analysis of where compositional role information is preserved and lost.
2. **Dongda Li et al.** “Correlation Is Not Selection: Validating Automatic Evaluators for Robot Policies” *Under review at ICLR 2027*. Evaluation of reward models and VLM judges against large-scale, double-blind human comparisons.

Research Computing Allocation

NSF ACCESS-CI Discover Allocation — Principal Investigator

2026

Efficient Evidence Bottlenecks for Compositional Vision-Language Understanding. Awarded an ACCESS-CI Discover research computing allocation to support experiments on efficient compositional vision-language models.

Technical Skills

Languages: Python, C/C++, TypeScript, Rust

Machine Learning: PyTorch, Transformers, reinforcement learning, multimodal evaluation

Agentic AI systems: LLM tool use, agent workflows, MCP, REST APIs

Systems: Linux, CUDA workflows, Git, HPC/cluster experimentation

Robotics: ROS, SLAM, motion planning, navigation, control

Product Engineering & Open Source

Mindwtr — Founder & Lead Engineer

- Founded Mindwtr and lead engineering for the cross-platform, open-source, local-first application, reaching **25,000+ users**, **~2,000 daily active users**, and **2,000+ GitHub stars**; designed synchronization, optional AI features, MCP/REST automation, release pipelines, and multi-platform deployment.

Industry and Teaching Experience

Robotics Engineer

Sep. 2022–Jul. 2023

Goertek Inc., Algorithm Group, R&D

Shanghai, China

- Worked on robotic perception and control, SLAM, motion planning, navigation, and trajectory planning for autonomous robotic systems.

Teaching Assistant

Aug. 2023–Present

Syracuse University

Syracuse, NY

- CPS 181 Introduction to Computing; CIS 454 Software Implementation; CIS 675 Design and Analysis of Algorithms.

Robotics Engineer, Beijing Zhonghangzhi Inc.

Jul. 2015–Jun. 2016

Robotic navigation and control.

Selected Publications and Patent

1. **Dongda Li**, Zhaoquan Gu, Yuexuan Wang, Changwei Ren, and Francis C.M. Lau. “One Model Packs Thousands of Items with Recurrent Conditional Query Learning.” *Knowledge-Based Systems*, 235:107683, 2022. Reinforcement-learning-based neural combinatorial optimization.
2. Zhaoquan Gu, **Dongda Li**, Nadra Guizani, Xiaojiang Du, and Zhihong Tian. “An Aerial-Computing-Assisted Architecture for Large-Scale Sensor Networks.” *IEEE Wireless Communications*, 28(5):43–49, 2021.
3. **Dongda Li** et al. “Adler: A Resilient, High-Performance and Energy-Efficient UAV-Enabled Sensor System.” HKU Technical Report TR-2018-01.
4. Tong Shen et al. “Alano: An Efficient Neighbor Discovery Algorithm in an Energy-Restricted Large-Scale Network.” *IEEE MASS*, 2018.
5. Yukin Wang, **Dongda Li**, and Bin Dong. “Variable-Speed Rotor Blade, Coaxial Unmanned Helicopter and Single-Rotor Unmanned Helicopter.” Granted China patent CN115610648B, 2024.

Selected Earlier Projects

- Collaborative analysis and autonomous decision technology for intelligent manufacturing big data, National Key R&D Program of China (2018–2021).
- Software-defined wireless sensor network system (2017–2018) and UAV-enabled sensor system (2017).
- [MBZIRC 2017 International Robotics Challenge](#); core member of the **champion** team.
- Autonomous takeoff and landing of an intelligent quadrotor, undergraduate thesis (2014–2015).

Service, Awards, and Languages

Service: Reviewer, JSAC (2022), IROS (2020), Journal of Contemporary Mathematics (2024)

Awards: [Core member of the MBZIRC 2017 champion team](#)

Languages: Chinese (native), English (IELTS 7.0)

Volunteer: Chinese translation, CS 285 Deep Reinforcement Learning